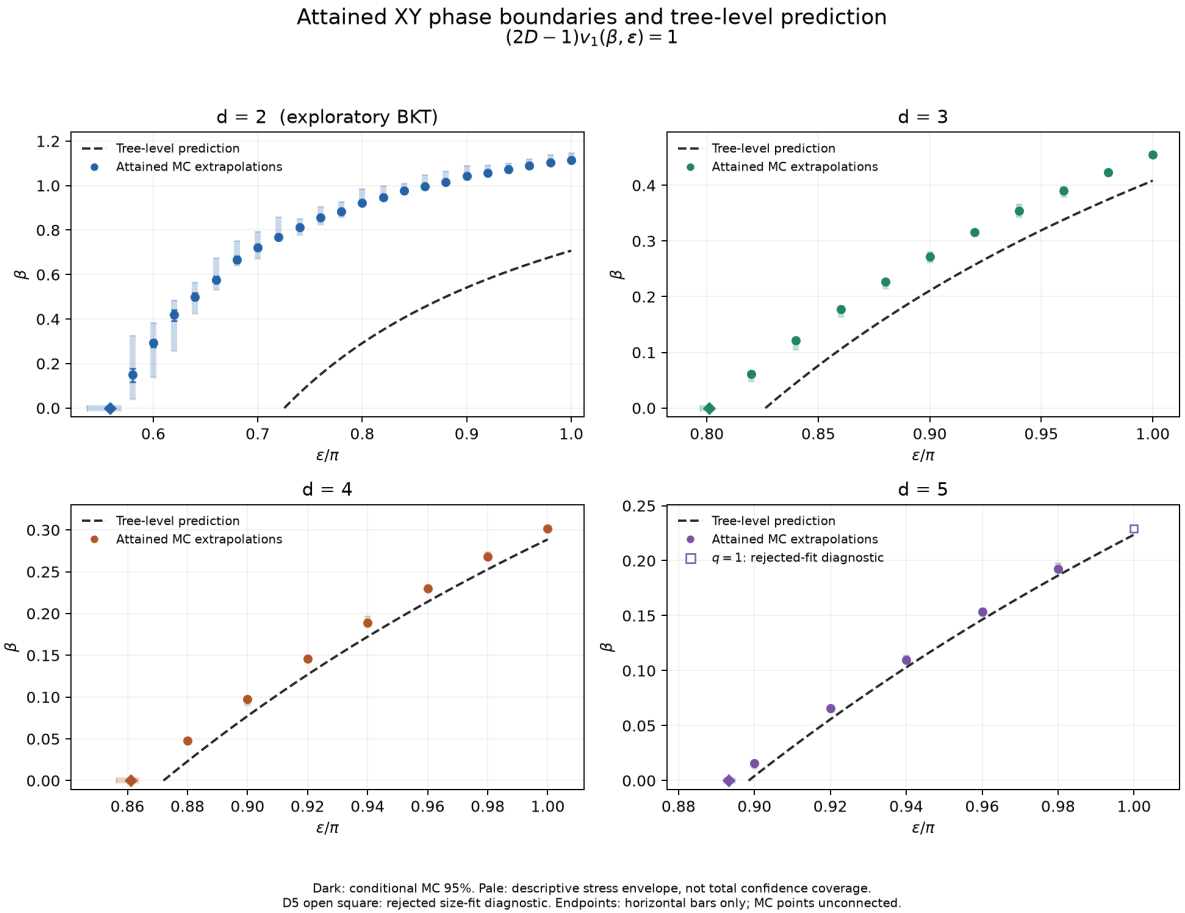
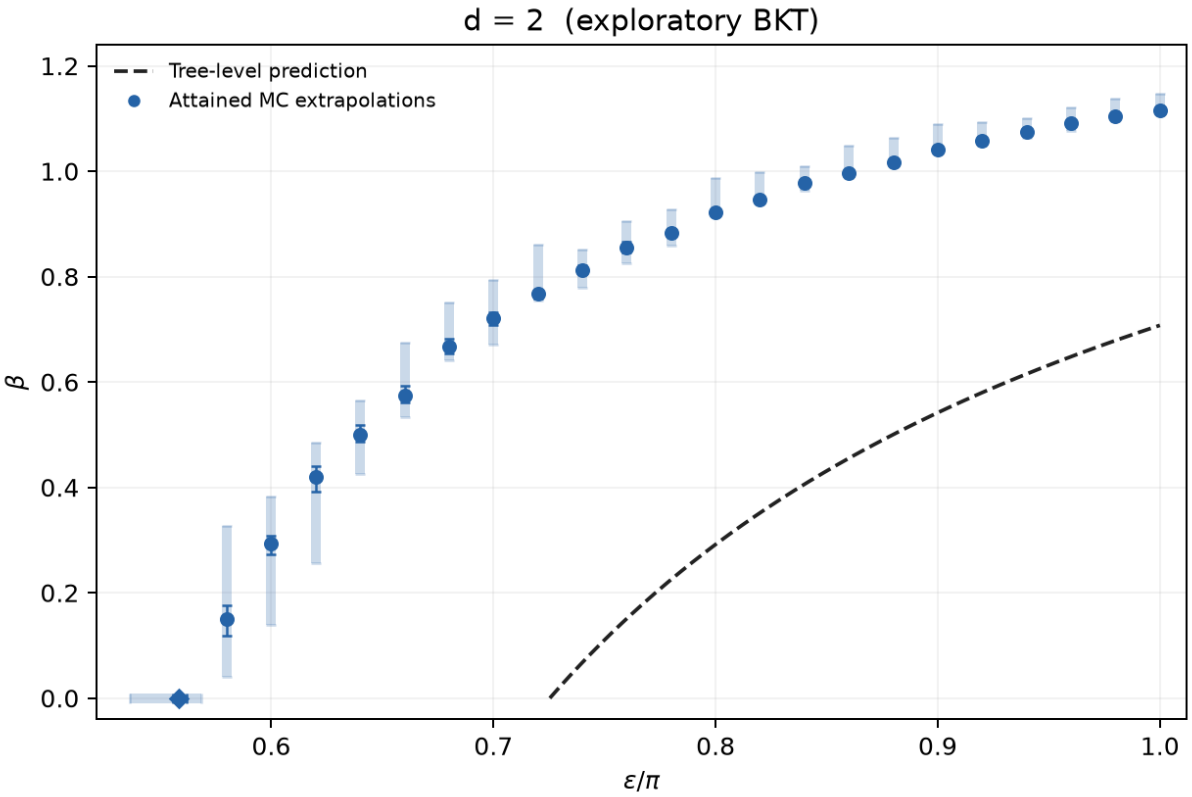
01 Phase boundaries: d = 2, 3, 4, 5



Unconnected MC extrapolations versus the full tree instability criterion (2d-1)v1=1. Dark bars: conditional MC95. Pale bars: descriptive sensitivity unions, not certified total confidence intervals. Beta=0 endpoints have horizontal bars; fixed-epsilon points have vertical bars. The open d=5 square at epsilon/pi=1 is a rejected-fit diagnostic, not an accepted critical coupling. Axes differ across panels.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

02 Phase boundary: d = 2

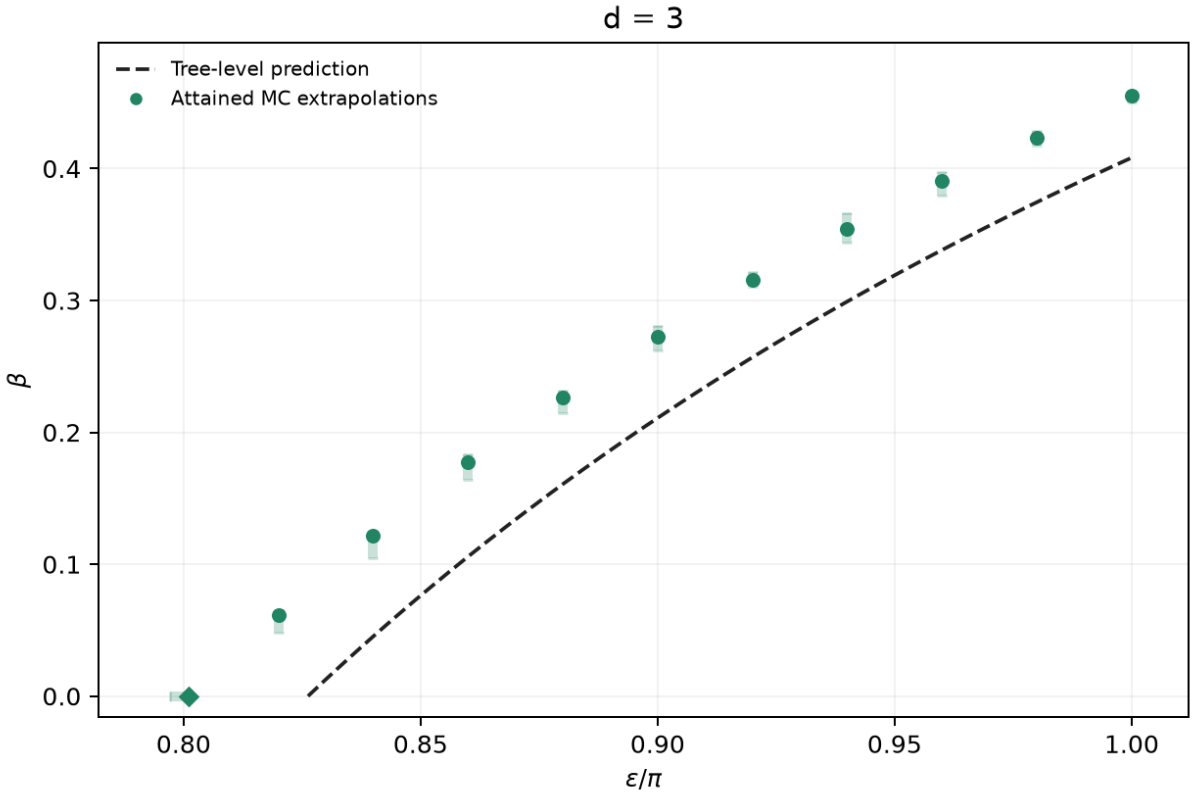


Dark: conditional MC 95%. Pale: descriptive stress envelope, not total confidence coverage.
D5 open square: rejected size-fit diagnostic. Endpoints: horizontal bars only; MC points unconnected.

Unconnected MC extrapolations versus the full tree criterion. Dark bars are conditional MC95; pale bars are descriptive sensitivity unions, not certified total confidence intervals. Beta=0 endpoints have horizontal bars; fixed-epsilon points have vertical bars. In d=2, N=512 is available only at representative slices/endpoints; absolute BKT accuracy is not certified.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

03 Phase boundary: d = 3

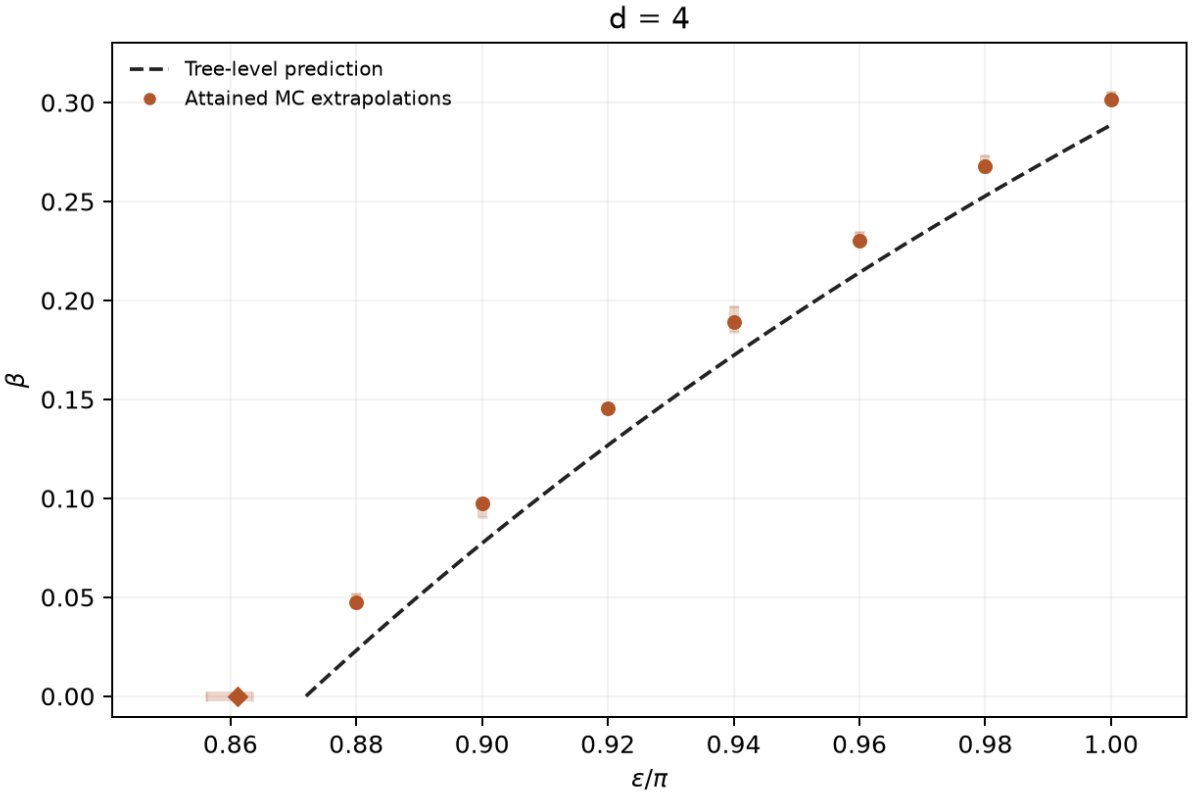


Dark: conditional MC 95%. Pale: descriptive stress envelope, not total confidence coverage.
D5 open square: rejected size-fit diagnostic. Endpoints: horizontal bars only; MC points unconnected.

Unconnected MC extrapolations versus the full tree criterion. Dark bars are conditional MC95; pale bars are descriptive sensitivity unions, not certified total confidence intervals. Beta=0 endpoints have horizontal bars; fixed-epsilon points have vertical bars.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

04 Phase boundary: d = 4

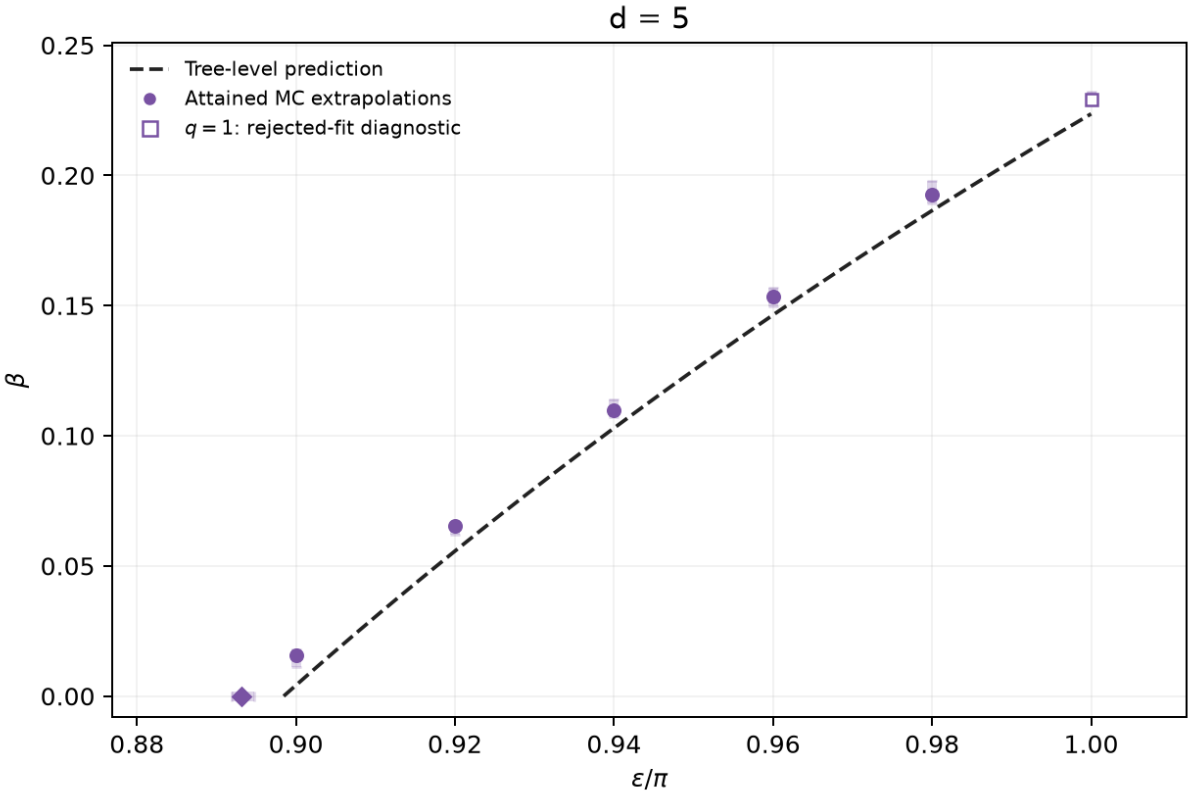


Dark: conditional MC 95%. Pale: descriptive stress envelope, not total confidence coverage.
D5 open square: rejected size-fit diagnostic. Endpoints: horizontal bars only; MC points unconnected.

Unconnected MC extrapolations versus the full tree criterion. Dark bars are conditional MC95; pale bars are descriptive sensitivity unions, not certified total confidence intervals. Beta=0 endpoints have horizontal bars; fixed-epsilon points have vertical bars.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

05 Phase boundary: d = 5

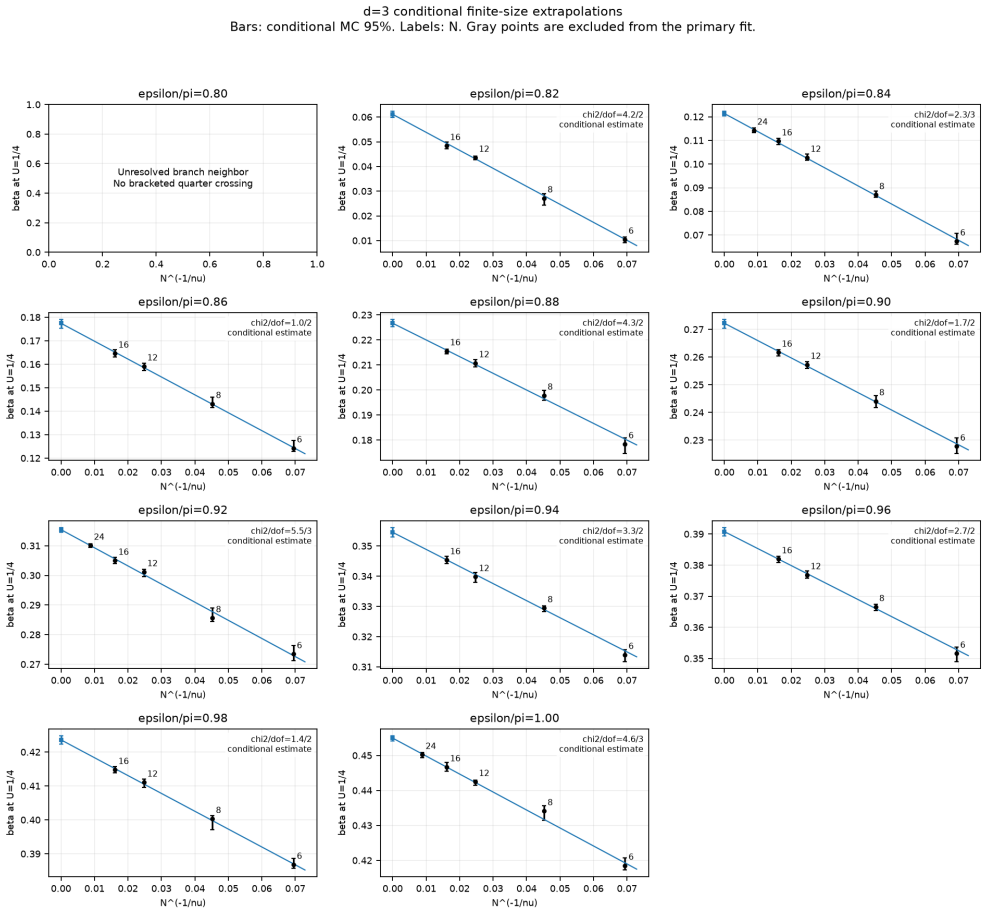


Dark: conditional MC 95%. Pale: descriptive stress envelope, not total confidence coverage.
D5 open square: rejected size-fit diagnostic. Endpoints: horizontal bars only; MC points unconnected.

Unconnected MC extrapolations versus the full tree criterion. Dark bars are conditional MC95; pale bars are descriptive sensitivity unions, not certified total confidence intervals. Beta=0 endpoints have horizontal bars; fixed-epsilon points have vertical bars. The open square at epsilon/pi=1 is a rejected-fit diagnostic, not an accepted critical coupling.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

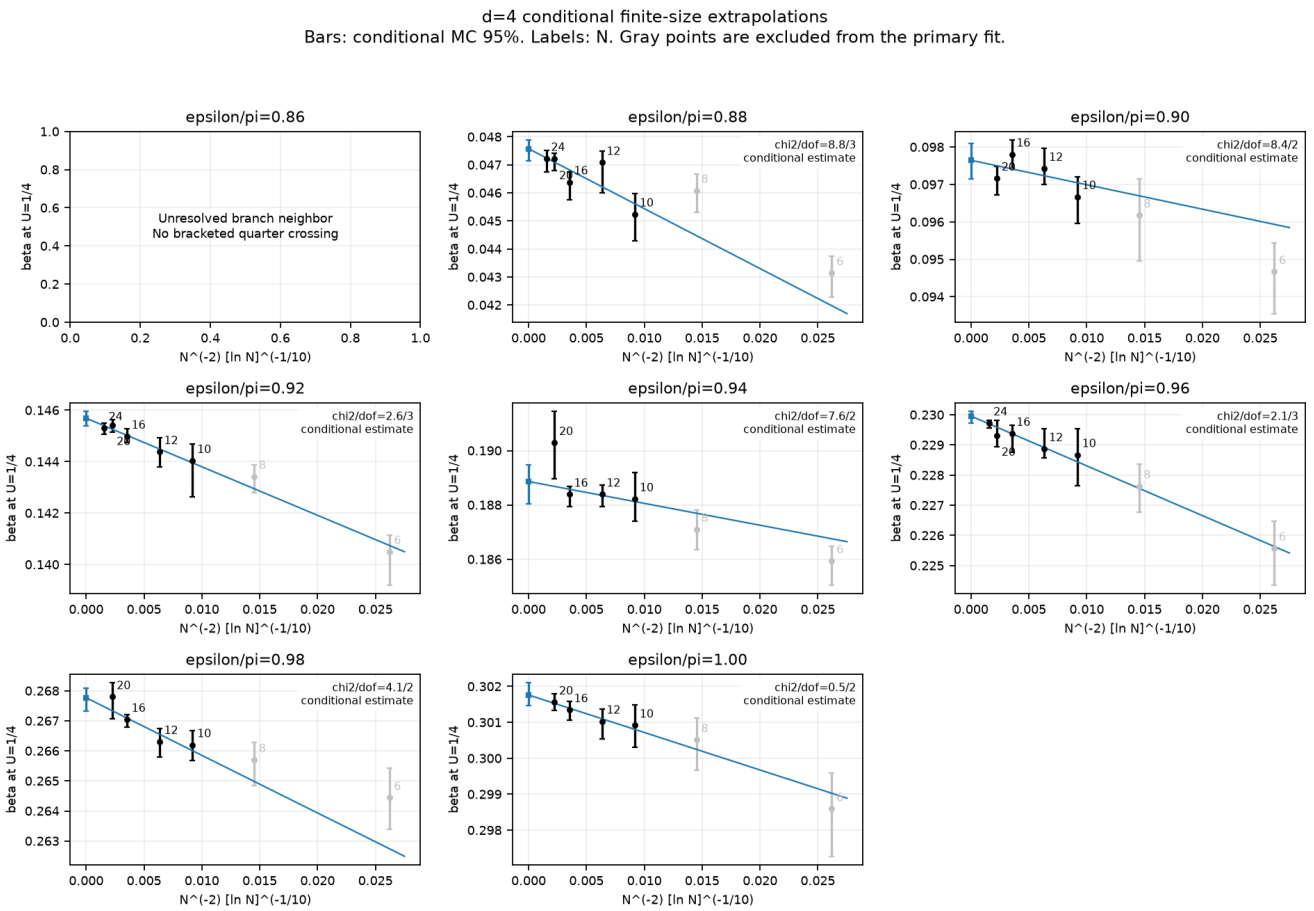
06 Finite-size transition fits: d = 3



Binder U=1/4 pseudo-critical locations, plotted against N^(-1/nu), nu=0.67169. Interpolation, size cuts and correction fits are sensitivity assumptions. This is the phase-boundary analysis, not the later common-xi scaling-dimension estimator.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

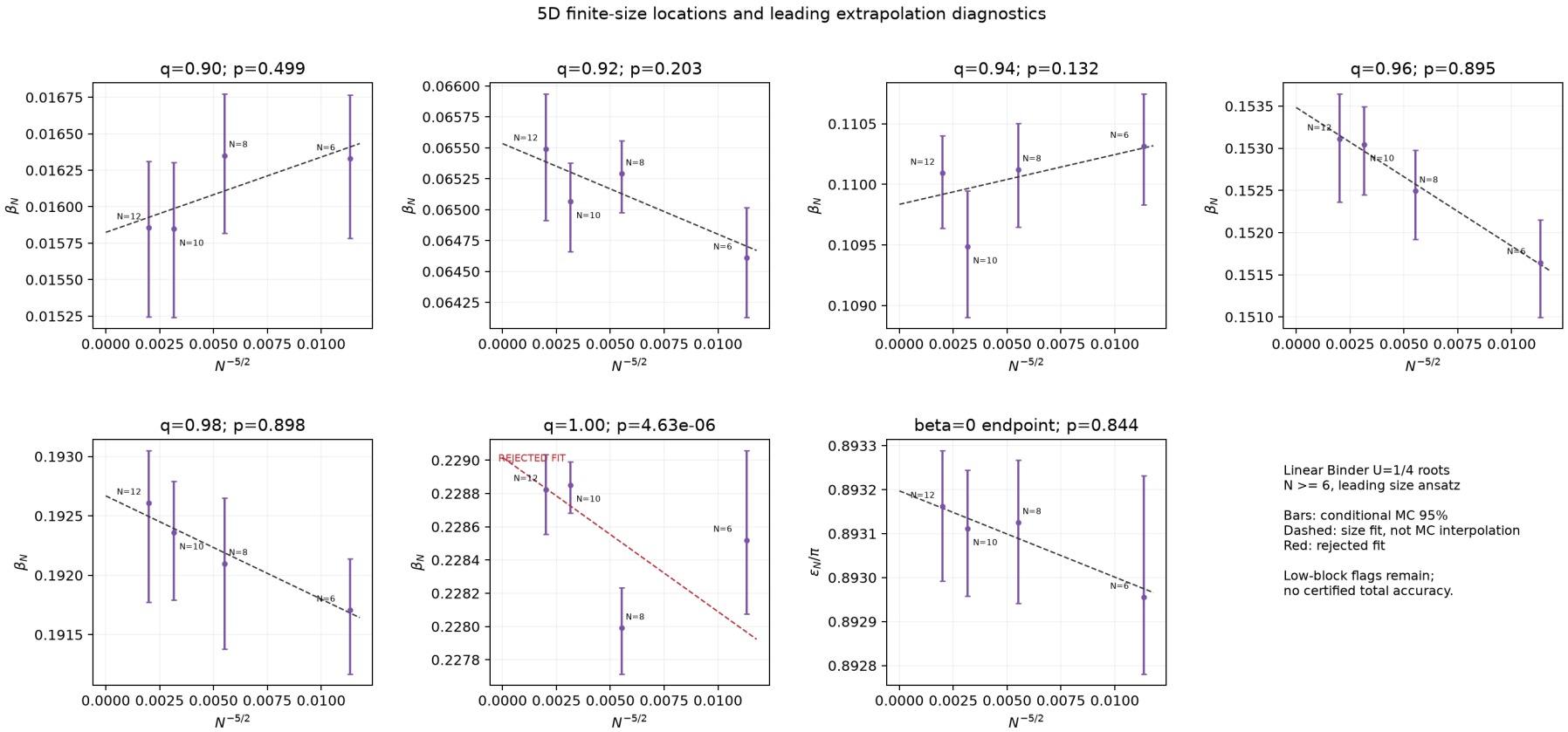
07 Finite-size transition fits: d = 4



Binder U=1/4 locations use the leading N^-2 (ln N)^(-1/10) shift coordinate. Uniform primary positive-beta cut N>=10 was chosen after inspecting pilot residuals. Fits and error coverage remain conditional.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

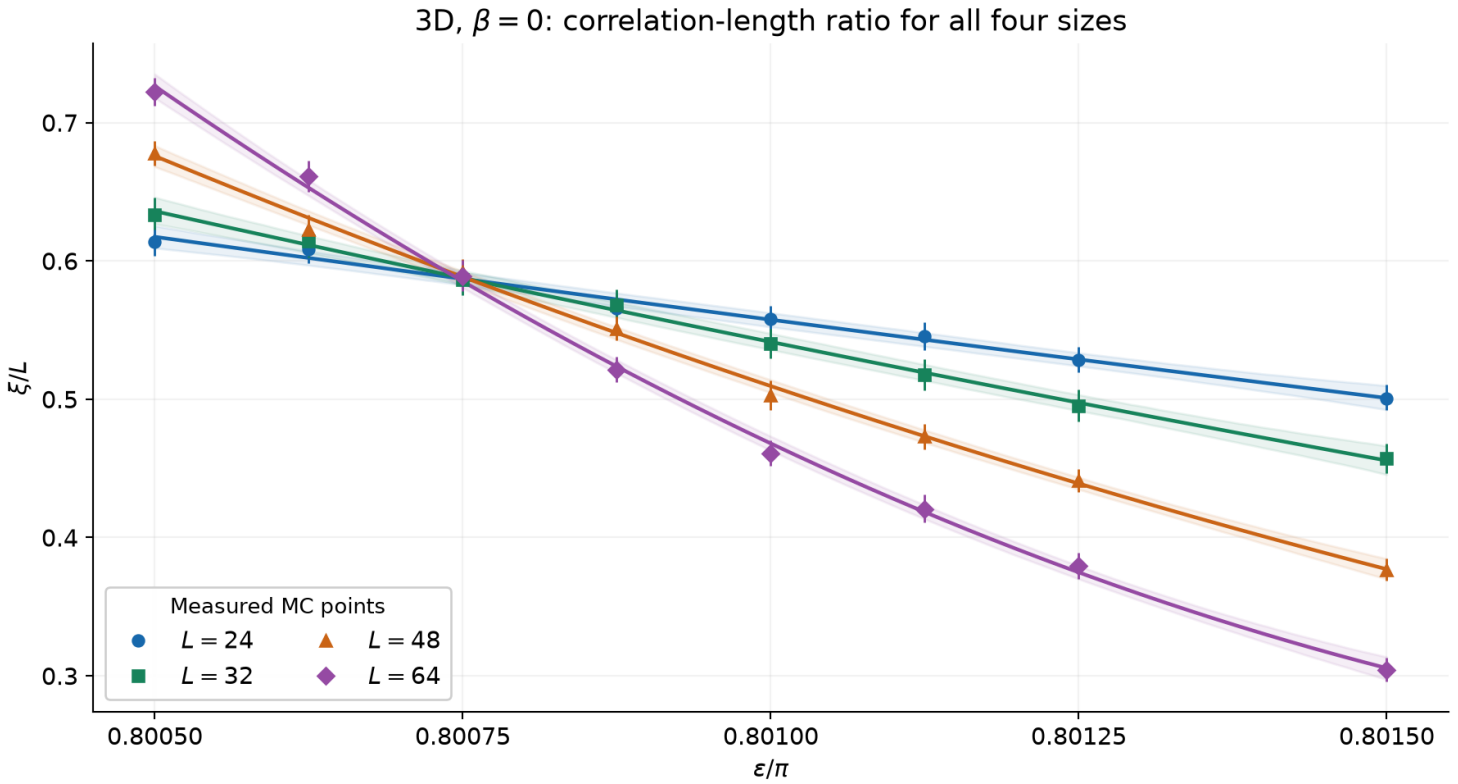
08 Finite-size transition diagnostics: d = 5



Dedicated above-upper-critical-dimension leading N^-5/2 shift, with additive N^-3 stress tests. Quarter-Binder is a fixed ordered-side criterion, not a universal finite-box critical value. Rejected fits and zero-degree-of-freedom diagnostics must not be presented as validated extrapolations.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

09 3D beta=0: four sizes on one plot

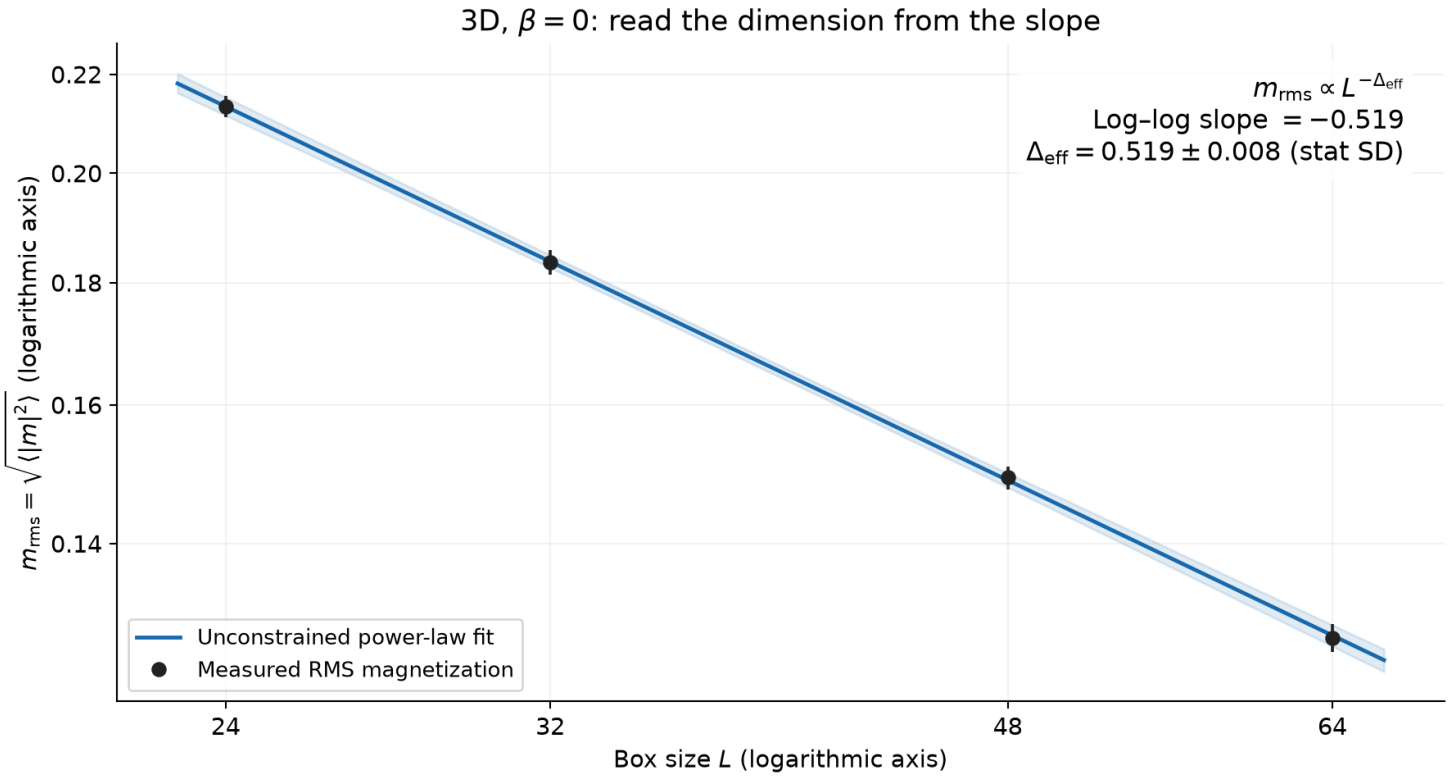


Four independent quadratic fits; no common crossing imposed.
Points and shaded bands: conditional MC 95%, excluding fit-model uncertainty.

Measured xi/L for L=24,32,48,64 with independent broad-window quadratic response fits. No common crossing or critical xi/L is imposed. Bars/bands are conditional MC95. Finite-size curves need not cross at exactly one epsilon.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

10 3D beta=0: reading off the spin dimension

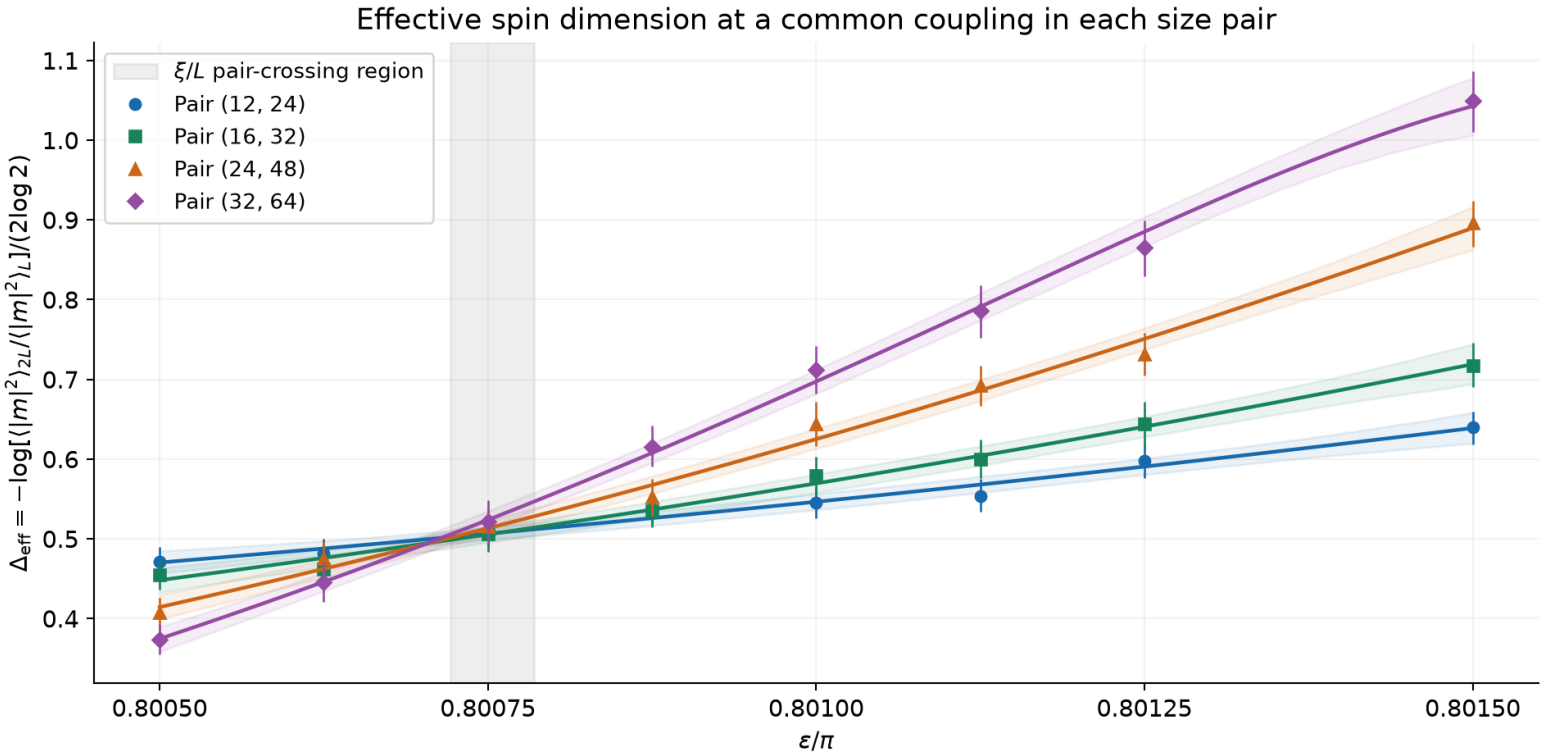


Fixed measured $\varepsilon/\pi = 0.80075$, near the ξ/L crossings; four sizes fitted together.
Bars/band: conditional MC 95%. Coupling-location and finite-size corrections are not included in this fit error.

At the exactly sampled epsilon/pi=0.80075, RMS magnetization scales approximately as L^-Delta_eff over L=24,32,48,64. Correlated fit gives Delta_eff=0.51942 +/- 0.00795 statistical SD. This is a representative fixed-epsilon finite-size slope, not a calibrated infinite-volume CFT result.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

11 3D beta=0: effective dimensions across the scan

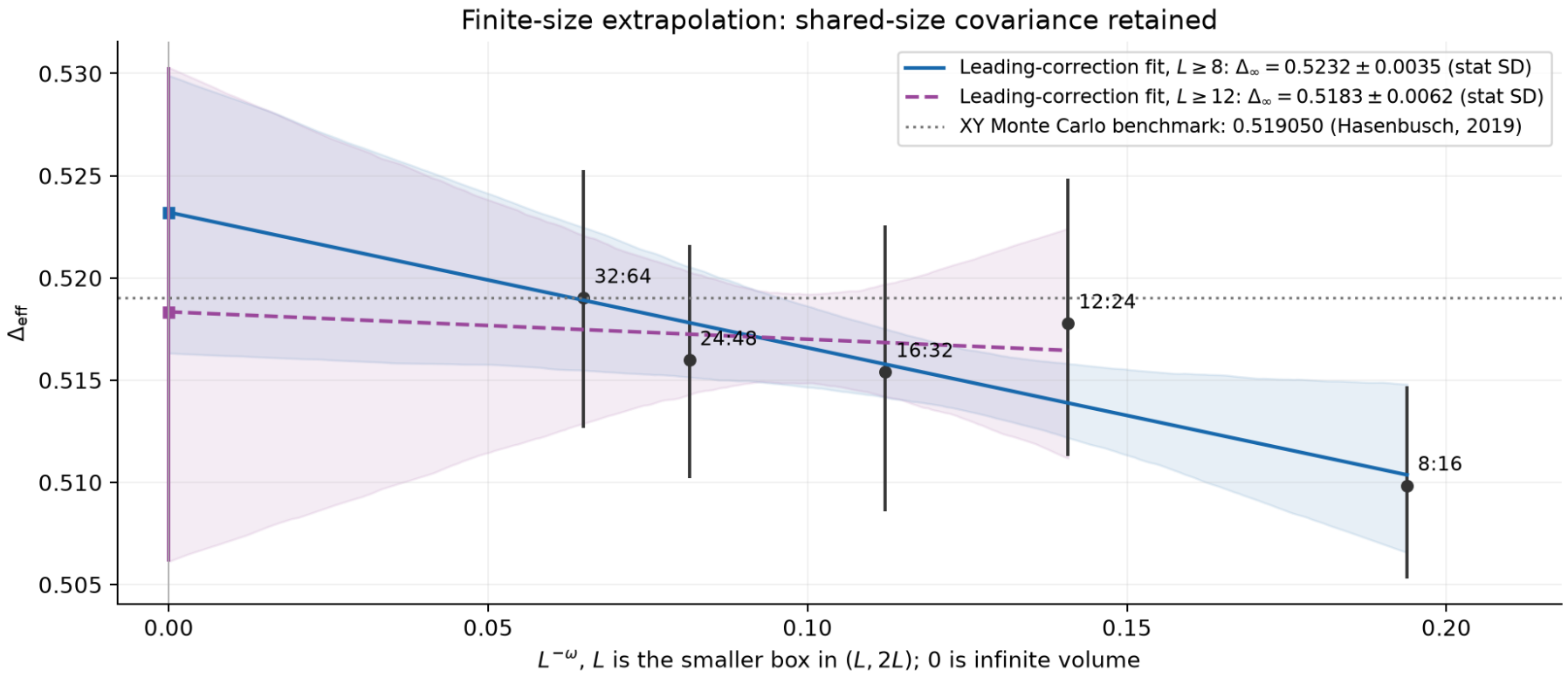


Points: direct measured ratios. Curves/bands: independent quadratic magnetization responses; conditional MC 95%.
Gray strip: descriptive union of two finite-size crossing intervals, not an infinite-volume critical-coupling interval.
Finite-size corrections remain; these effective-dimension curves need not cross at one exact point.

Delta_eff(L,2L;q)=-ln(_2L/_L)/(2 ln2), evaluated at common q. Four overlapping size pairs retain shared-source covariance. Grey band is a descriptive union of finite-pair xi/L crossing intervals, not a thermodynamic epsilon_c confidence interval.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.

12 3D beta=0: conditional size extrapolations

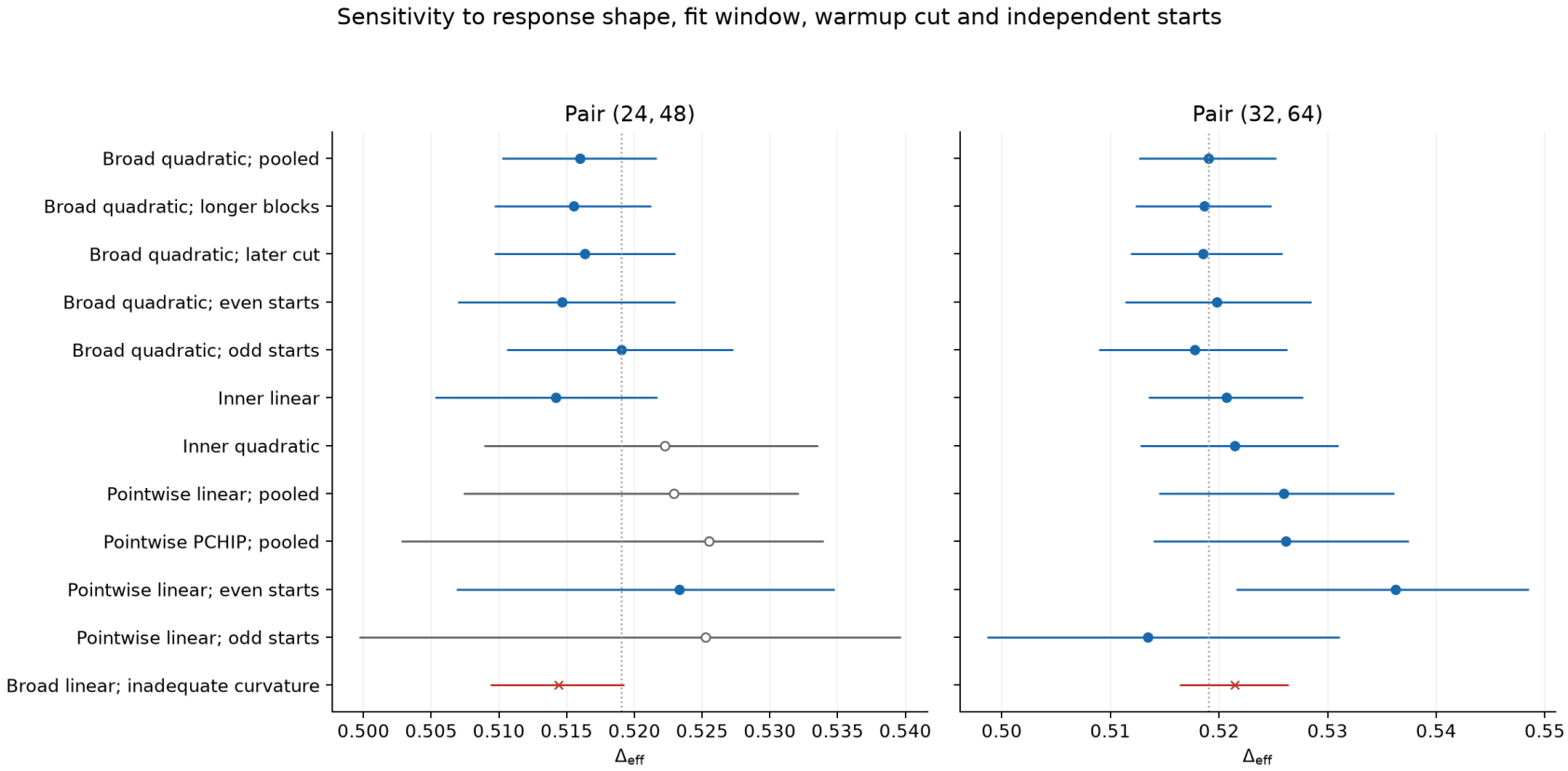


Bars/bands: conditional MC 95%; fixed $\omega = 0.789$ is WF-informed. Size cuts are alternatives, not independent measurements. 6 : 12 is unresolved.

Correlated extrapolations of common-xi quotient dimensions with fixed Wilson-Fisher-informed correction exponent omega=0.789. These illustrate model and size-cut sensitivity; no final Delta +/- stat +/- sys is established. Low/missing-validity pairs are not silently accepted.

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13 3D beta=0: analysis sensitivity



Bars: conditional MC 95%. Open circles: unresolved roots; red crosses: rejected response shape. Rows reuse correlated data.
Unresolved intervals use surviving draws. Dotted line: XY Monte Carlo benchmark, 0.519050 (Hasenbusch, 2019).

Local response model/window, late-cut, block-length and independent-start sensitivity. A stable selected family is not proof of complete mixing or total uncertainty coverage; rejected responses and unresolved alternatives are retained in the source analysis.

Use the companion PDF or SVG for vector graphics; PNG for quick slide insertion.